MARK SCHEME · TEACHER COPY

10BR · Chapter 3: Hardware

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 75 marks

Question 1 · May/June 2023 Paper 11, original Q3 (a),(b) [4 marks]

Question	Answer	Marks
1(a)	**One** mark for each correctly circled storage device: • Compact disk (CD) • Solid-state drive (SSD) • Hard disk drive (HDD)	3
1(b))	• C	1

Question 2 · May/June 2023 Paper 13, original Q9 (a),(b),(c) [7 marks]

Question	Answer	Marks
2(a)	Any **one** from: • They can both be used to identify a device (on a network) • They can both be static / dynamic • They are both unique (to a device on a network) • They can both be assigned by a router • They can both be public/private	1

Total: 75 marks

2(b)	**Four** from: • IPv4 is usually written as denary • … IPv6 usually written as hexadecimal • IPv4 is separated using dots • … Pv6 is separated using colons • IPv4 is 32-bit • … IPv6 is 128-bit • IPv4 is 4 groups of digits • … IPv6 is 8 groups of digits • IPv4 digits are between 0 and 255 • … IPv6 digits are between 0000 and FFFF • IPv4 all 0s are displayed • … IPv6 can use double colons to replace repeated groups of 0000 • IPv4 has fewer available unique addresses • ... IPv6 has more available unique addresses	**4**
2(c)(i)	• Domain name server // DNS	**1**
2(c)(ii)	• Web browser	**1**

Question 3 · Oct/Nov 2023 Paper 12, original Q3 (a),(b),(c) [6 marks]

Question	Answer	Marks
3(a)(i)	– The maximum number of FDE cycles/instructions a CPU can perform/process/execute in a second	**1**
3(a)(ii)	– Increases/improves the performance // Tasks can be performed quicker/faster – … because more FDE cycles/instructions can be processed in a second	**2**

3(b)	– **Stores** addresses … – … of next instruction/data to be fetched // where data is to be written to	**2**
3(c)	– Instruction set	**1**

Question 4 · May/June 2024 Paper 12, original Q9 (a),(b) [8 marks]

Question	Answer	Marks
4(a)	Any **two** from: • To store data/files **permanently** • … otherwise, data/programs would need to be downloaded/entered/installed every time the computer was used • To allow **software** to be **installed** on the computer • For the creation of virtual memory	**2**

4(b)	Any **six** from (**Max 3** for one type only): • Solid-state has no moving parts • … optical does have moving parts • Solid-state is faster to **access** data • … optical is slower to **access** data • Solid-state uses less power to operate • … optical uses more power to operate • Solid-state runs quietly • … optical makes more noise • Solid state is more durable/robust • … optical is less durable/robust • Solid state storage has a large storage capacity • … optical has a much smaller limit to its storage capacity • Solid-state is more expensive **per GB** of data • … optical is cheaper **per GB** of data • Solid-state stores data onto silicon chips/transistors using logic (floating/control/NAND/NOR) gates to control the movement of electrons • … optical stores data by using lasers to burning pits and lands onto a disk	**6**

Question 5 · Oct/Nov 2024 Paper 11, original Q5 (a),(b),(c),(d),(e) [11 marks]

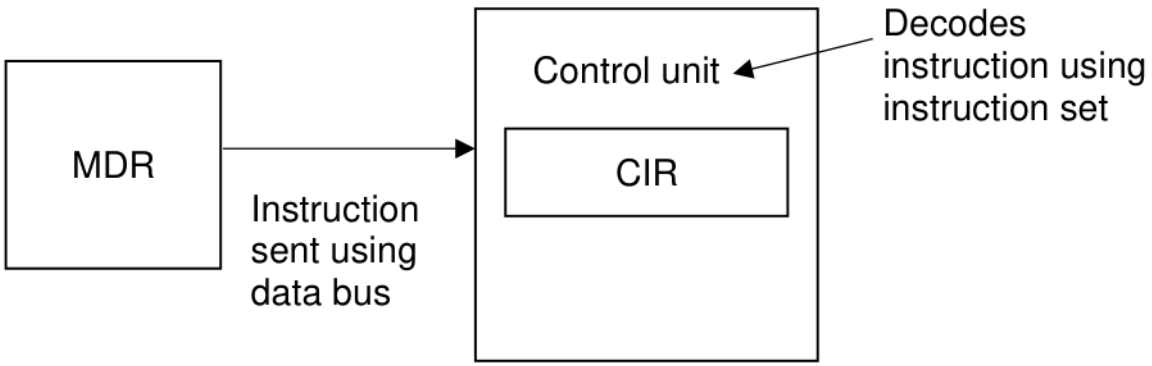
Question	Answer	Marks
5(a)	Any **two** from: • Program counter // PC • Memory address register // MAR • Memory data register // MDR • Current instruction register // CIR	**2**

Total: 75 marks

5(b)	Any **three** from: • **CIR/CU** receives the instruction from the **MDR** // Instruction sent from **MDR** to **CIR/CU** • … using the data bus • Instruction is **separated** into opcode and operand • Control unit decodes the **instruction** • … using an instruction set	**3**
5(c)	Any **one** from: • Accumulator • Memory address register // MAR • Memory data register // MDR	**1**
5(d)	• data • address • control	**3**
5(e)	• It can now execute **more** instructions/FDE **per second** • … this will increase the performance of the CPU	**2**

Question 6 · Oct/Nov 2024 Paper 13, original Q3 (a),(b) [6 marks]

Question	Answer	Marks
6(a)	Any **two** from: • Program counter // PC • Memory address register // MAR • Current instruction register // CIR	**2**

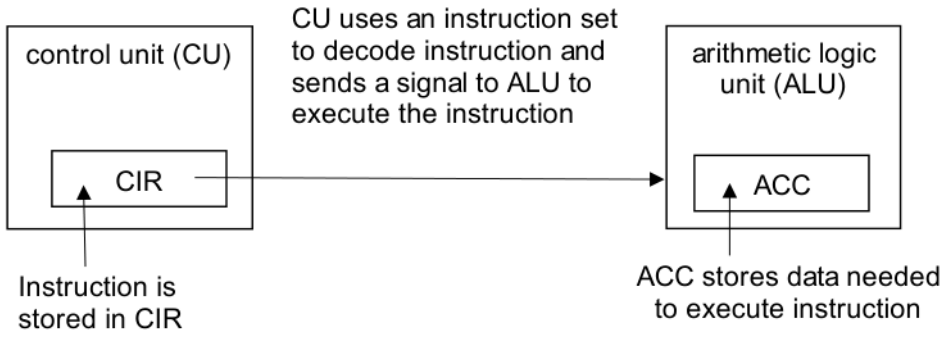
6(b)	**One** mark for each correct part of the diagram. The diagram shows: • Data/Instruction sent from the MDR to the CIR/CU • … using the data bus • … the CIR that is built into the CU • Data/instruction separated into operand and op code • Control unit/CU decodes instruction • … using an instruction set For example:  	**4**

Question 7 · May/June 2025 Paper 11, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks
7(a)	Any **one** from: • Read only memory // ROM • Cache	**1**

7(b)	Any **three** from: • RAM is directly accessed by the CPU. • … whereas secondary storage is not directly accessed by the CPU. • All secondary storage is non-volatile storage. • … whereas RAM is volatile storage.	**3**
7(c)	Any **three** from: • Virtual memory can act as an extension to the RAM . • The secondary storage device is **partitioned** to create the virtual memory. • Pages can be **transferred** from the RAM to the **virtual memory** … • … to stop the RAM getting full // … when the RAM is full.	**3**
7(d)(i)	Any **two** from: • It has no moving parts. • It uses semiconductor chips. • It uses NAND/NOR/Flip-flops technology. • It uses cells and transistors (that are laid out in a grid). • It uses control gates and floating gates. • **Stores** data as **electrical** charges.	**2**
7(d)(ii)	Any **two** from: • Solid-state drive // SSD • USB flash memory drive • SD card	**2**

Question 8 · Oct/Nov 2025 Paper 12, original Q4 (a),(b),(c) [11 marks]

Question	Answer	Marks
8(a)	**One** mark for each correct part of the diagram. The diagram shows: • CIR is built into CU • CIR stores instruction • CU decodes the instruction • Instruction is decoded using an instruction set // Instruction is separated into opcode and operand • Signal is sent from CIR to ALU (to execute) • CU sends signal to RAM to fetch any data needed for instruction • ACC is built into ALU • ACC stores any data needed for instruction • ALU executes the instruction // ALU performs the mathematic/logical operation (needed by the instruction) Example: 	5
8(b)	Any **two** from: • Independent/self-contained **processing** unit • It executes/processes **instructions** // It performs the FDE cycle • It contains registers/units (to process data)	2

8(c)	**Four** from (MAX 2 if no expansion given): • The number of cores could be increased // by example … • … this means that more instructions can be processed **simultaneously** • The clock speed could be increased // by example … • … this means that more instructions can be processed (by a core) **per second** • The cache could be increased // by example … • … this means that more frequently used data/instructions can be stored (to be accessed faster than RAM)	**4**

Question 9 · May/June 2026 Paper 11, original Q2 (a),(b),(c) [11 marks]

Question	Answer	Marks	Guidance
9	Examiners must award **whole marks** (not half marks, or other fractions).		
9(a)(i)	**One** from: • Program counter // PC • Memory Address register // MAR	**1**	Mark the first answer given.
9(a)(ii)	**Three** from: • The instruction is sent/copied to the current instruction register/CIR/control unit/CU // The CIR/CU receives the instruction // The CIR stores/holds the instruction … • … from the memory data register/MDR • The **control unit/CU decodes** the instruction • An instruction set is used (to decode the instruction) • The instruction is decoded/converted into the Opcode and Operand	**3**	BOD data in place of instruction MP1 – BOD instruction register/IR in place of CIR MP1 – this is given for the instruction being sent to the CIR/control unit. Allow if it is coming from an incorrect or inaccurate place e.g. MAR/PC MP3 – It must clearly state the CU is decoding the instruction

9(a)(iii)	**One** from: • Arithmetic logic unit // ALU • Accumulator // ACC • Control unit // CU • (Data/address/control) bus • Memory data register // MDR • Memory address register // MAR • Program counter // PC • Random access memory // RAM	**1**	Mark the first answer given. Register alone is NE Do not award CIR Award appropriate registers that are outside the specification e.g. Index register (IX) and status register (SR)
9(b)	**Two** from: • It can execute **more** instructions/FDE/FE/clock cycles **simultaneously** // **four** instructions/FDE/FE/clock cycles can be executed **simultaneously** rather than **two** • It can execute **more** instructions/FDE/FE/clock cycles **per second** • … as it has **double the number** of cores // It has **twice as many** cores // It has **more** cores // It has **4** cores instead of **2** cores	**2**	Tasks/processes/code/information/commands/actions/data is TV for instructions MP1 and 2 – BOD clock ticks/pulses in place of cycles
9(c)(i)	**Two** from: For example: • Touchscreen • Microphone • Keyboard/keypad • Button • Sensor • (Digital) camera	**2**	Mark the first answer given. Award any appropriate example of a sensor. Sensor can only be awarded once. Do not allow onscreen keyboard as this is software. Do not allow touchpad as it is not relevant to context.
9(c)(ii)	**Two** from: For example: • Screen // touchscreen • Speaker • LED // light • Actuator	**2**	Mark the first answer given. Display alone is NE for screen.